

Task 58: Max Area of Island

Problem Description

You are given an $m \times n$ binary matrix grid. An island is a group of 1's (representing land) connected 4-directionally (horizontal or vertical.) You may assume all four edges of the grid are surrounded by water. The area of an island is the number of cells with a value 1 in the island. Return the maximum area of an island in grid. If there is no island, return 0.

Java Solution

```
class Solution {
    public int maxAreaOfIsland(int[][] grid) {
        int maxArea = 0;
        for (int i = 0; i < grid.length; i++) {
            for (int j = 0; j < grid[0].length; j++) {
                if (grid[i][j] == 1) {
                    maxArea = Math.max(maxArea, dfs(grid, i, j));
                }
            }
        }
        return maxArea;
    }

    private int dfs(int[][] grid, int i, int j) {
        if (i < 0 || i >= grid.length || j < 0 || j >= grid[0].length ||
            grid[i][j] == 0) {
            return 0;
        }
        grid[i][j] = 0;
        return 1 + dfs(grid, i + 1, j) + dfs(grid, i - 1, j) + dfs(grid,
            i, j + 1) + dfs(grid, i, j - 1);
    }
}
```